



CONTRACT INTELLIGENCE · AI / RAG · AZURE-NATIVE

Contract360

A retrieval-augmented assistant that answers natural-language questions over energy & infrastructure contracts
— with grounded, citation-backed answers.

Presented to Senior Leadership

Problem, Solution & Where It Creates Value

The problem

- **Long & high-stakes.** EPC, O&M and PPA contracts run to hundreds of pages of dense legal text.
- **Slow, manual review.** Finding obligations, deadlines, payment, termination & indemnity clauses is hours of work.
- **Generic AI can't be trusted.** Off-the-shelf chatbots hallucinate and give no citations — not auditable.

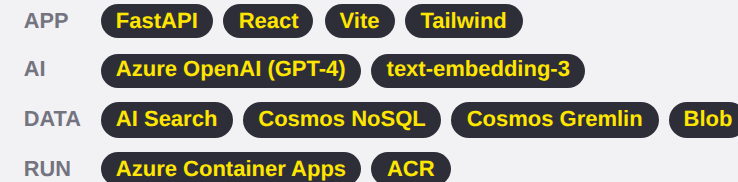
The solution — Contract360

- **Ask in plain English.** Upload a contract; query it conversationally — one contract or a portfolio.
- **Grounded & cited.** Every answer is drawn from real clause text and cites title, page range & source.
- **Contract-aware.** Clause-aware chunking, a hierarchical tree and a legal knowledge graph.

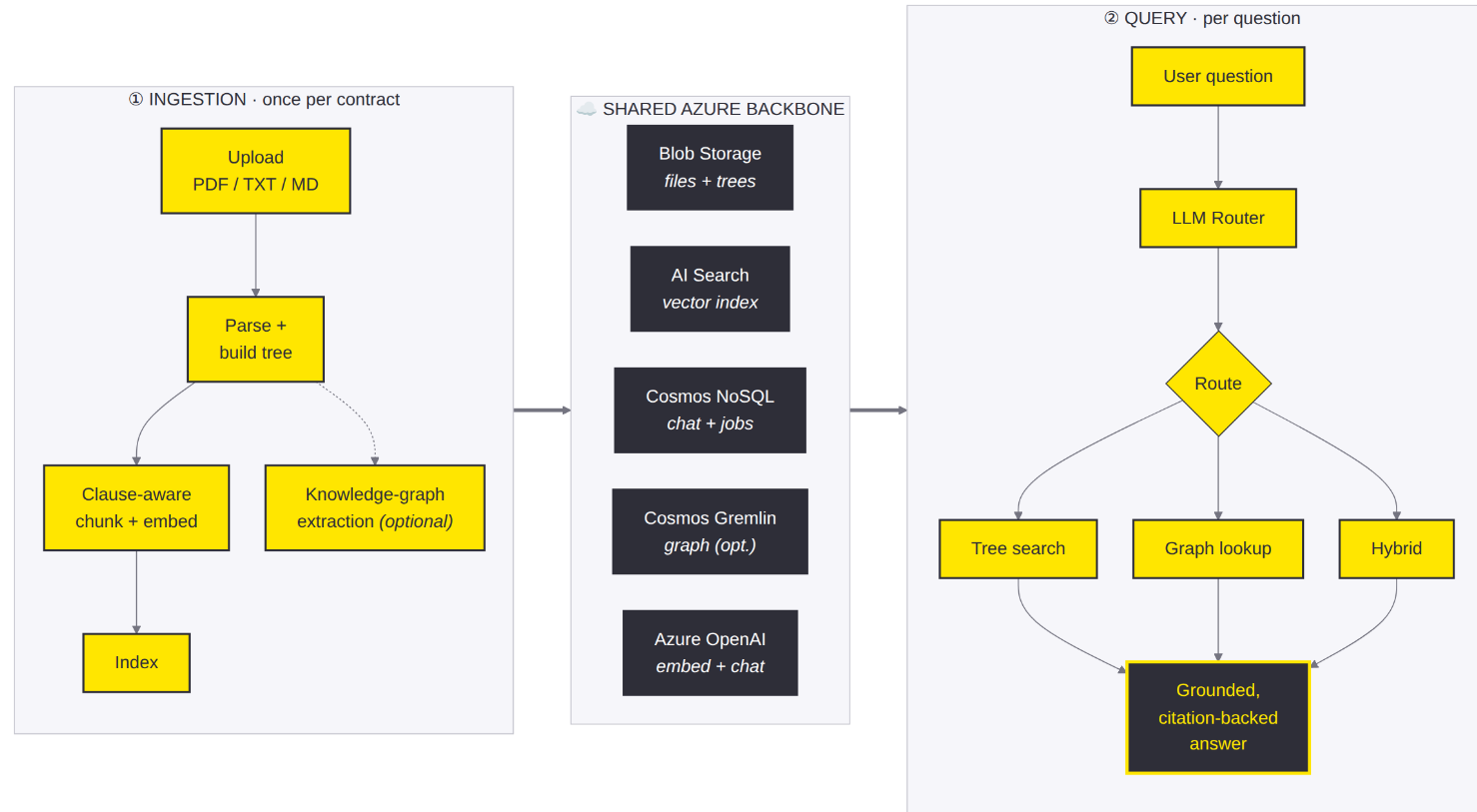
Business value

- **Faster review & due diligence.** Cut manual reading time for legal, commercial & deal teams.
- **Audit-ready answers.** Citation trail supports compliance and risk decisions.
- **Portfolio insight.** Surface obligations, rights & deadlines across many contracts at once.

Technology stack



High-Level Architecture

**INGEST ONCE**

Parse → tree → clause chunks + embeddings → searchable index.

ROUTE EACH QUERY

An LLM router picks tree, graph or hybrid retrieval per question.

ANSWER, GROUNDED

Azure OpenAI answers only from retrieved context — with citations.

Detailed Architecture

FIVE LAYERS

Frontend

React + TypeScript SPA — chat, sidebar, upload.

FastAPI

Sessions, ask (sync/stream), async ingest.

Query pipeline

LLM router → search / graph / hybrid / tree → AnswerGenerator.

Ingestion pipeline

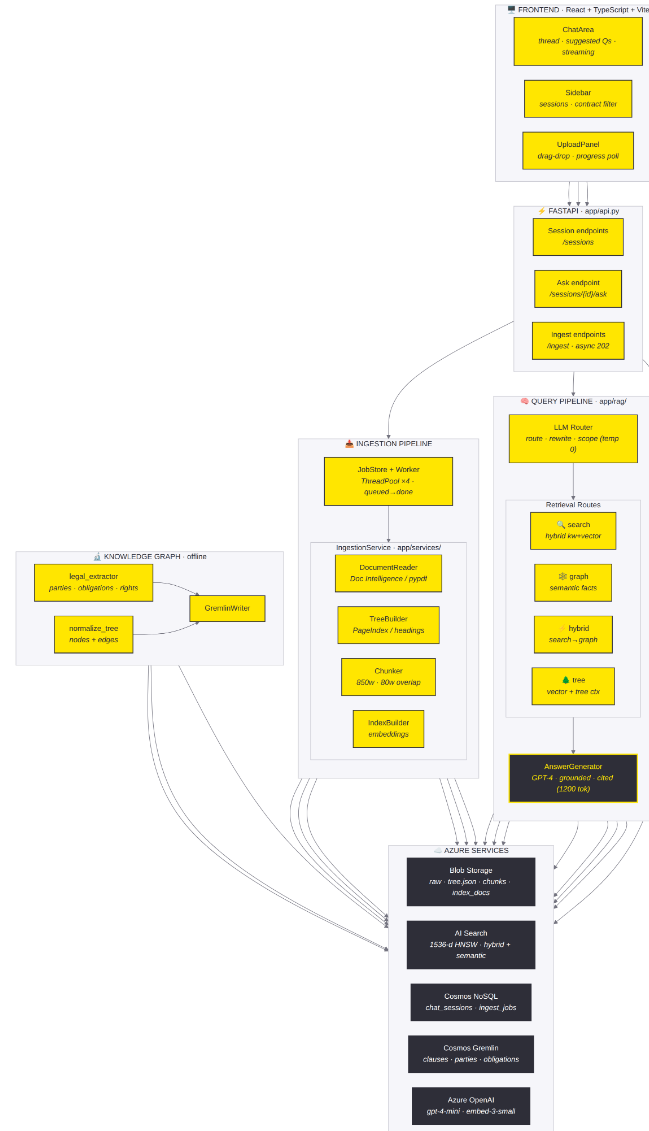
Worker pool: read → tree → chunk → embed → index.

Knowledge graph

Offline: parties, obligations, rights → Gremlin.

Azure services

Blob · AI Search · Cosmos NoSQL · Gremlin · OpenAI.



DESIGN HIGHLIGHTS

Async ingestion

Upload returns immediately (202); a thread pool processes files with live progress.

Pluggable parsing

Azure Document Intelligence with a pypdf fallback.

Cited answers

Sources carried end-to-end into every response.

Scales out

Container Apps; Service Bus path for multi-instance.

Config-driven

Graph & Doc-Intelligence toggle on/off via env.

See It In Action

Live demo flow

- **1. Upload a contract.** Drag-drop a PDF; watch parse → embed → index progress live.
- **2. Ask a question.** “What are the contractor's payment obligations?”
- **3. Read a grounded answer.** Response streams in with inline clause citations.
- **4. Show the routing.** A relationship question (graph) vs. a lookup (tree).
- **5. Multi-contract scope.** Filter to one contract or query across several.

Sample questions to demo

- “Summarise the termination clauses and any notice periods.”
- “What deadlines does the contractor have in the first 90 days?”
- “Which party bears liability for delay, and is there a cap?”
- “List the indemnities granted to the owner.”

Backup plan

A pre-recorded demo video is ready in case of live connectivity or environment issues.

[insert link / embed before the meeting]

What to watch for

- **Speed.** Answer in seconds; ingestion runs in the background.
- **Trust.** Each claim links to a clause — open it to verify.
- **Range.** One assistant handles lookup, summary & relationship questions.

Roadmap (post-MVP)

- Content-aware scoping by party / contract.
- Reasoning-based navigation of the contract tree.
- Knowledge-graph portfolio analytics across contracts.
- Azure Entra ID authentication for production.